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Memory Retention Experiment: Factorial Analysis of Typing vs. Writing

A REPORT

SUBMITTED BY

Koushan Saha (MD2513)

UNDER THE GUIDANCE OF

Advisor
Prof. Kiranmoy Das

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1 Experiment Overview

The experiment investigates memory retention as a function of note-taking modality and related cognitive factors. Each participant is presented with a list of 20 words and subsequently tested for recall after a fixed time delay. The **response variable** is the number of words correctly recalled out of 20.

2 Factors and Levels

Factor A: Method (Writing vs. Typing)

The primary factor of interest is the **mode of transcription**. Participants either write the word list by hand (0) or type it (1). Handwriting is a slower, more deliberate process that encourages deeper *elaborative encoding*, and is therefore expected to produce higher recall on average — though its advantage is likely conditional on word difficulty and transcription pace.

Factor B: Time (Slow vs. Fast)

The second factor captures the **pace of transcription**. Participants either work deliberately with more time (0) (around 4.5 minutes) or with less time (1) (around 2.5 minutes). A slower pace allows more time per word for processing and rehearsal, so the **Slow** condition is expected to benefit recall.

Factor C: Difficulty (Easy vs. Hard)

The third factor manipulates the **linguistic difficulty of the word list**, ranging from common high-frequency words (0) to uncommon, abstract, or low-frequency words (1). Hard words are more difficult to associate with existing memory structures and are therefore expected to reduce recall; however, the direction of the interaction with Method is not immediately obvious, since handwriting may help more — or no more — with unfamiliar vocabulary depending on prior semantic knowledge.

Factor D: Mental State (Fresh vs. Tired)

The fourth factor represents the **cognitive state of the participant** at the time of the experiment, either mentally fresh (0) or mentally tired (1). Fatigue is well documented to impair working memory capacity and encoding efficiency, so the **Fresh** condition is expected to produce higher recall; additionally, fatigue may compound the adverse effect of hard words, since both act simultaneously against the cognitive resources needed for retention.

3 Critical Assumption: Equal Memory Capacity Across Experimental Units

All experimental units are assumed to possess equal baseline memory capacity; that is, individual differences in cognitive ability or working memory span are treated as a **nuisance variable** that cannot be controlled, and are assumed uncorrelated with both the error term and the assigned treatment:

$$\mathbb{E}[\varepsilon_{ijklm} \mid \text{memory capacity of unit } m] = 0.$$

4 Choosing Full 2^4 Factorial Design

With 32 experimental units available, we are doing 2^4 factorial design, with 2 replications.

5 Experimental Controls and Randomization of Stimuli

A unique list of 20 words was generated for each of the 32 experimental units to ensure generalizability across word difficulty levels, prevent confounding from mnemonic or phonetic features of any single list, and preserve the i.i.d. assumption $\varepsilon_{ijklm} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$ required by both fitted models.

6 Linear Model

6.1 Second-Order Model

The second-order linear model, which forms the primary estimand of this experiment, includes all main effects and two-factor interactions:

$$Y_{ijklm} = \mu + \alpha_i + \beta_j + \gamma_k + \delta_l + (\alpha\beta)_{ij} + (\alpha\gamma)_{ik} + (\alpha\delta)_{il} + (\beta\gamma)_{jk} + (\beta\delta)_{jl} + (\gamma\delta)_{kl} + \varepsilon_{ijklm},$$

where $i, j, k, l \in \{0, 1\}$ index the factor levels, $m \in \{1, 2\}$ indexes the replication, and $\varepsilon_{ijklm} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$. This model contains $1 + \binom{4}{1} + \binom{4}{2} = 1 + 4 + 6 = 11$ parameters (excluding σ^2).

6.2 Full Fourth-Order Model

The full saturated linear model includes every estimable effect in the 2^4 design — all main effects, all two-factor, three-factor, and the single four-factor interaction. Since the full 2^4 factorial with 2 replications is used, all $2^4 - 1 = 15$ effects are independently estimable without any aliasing:

$$Y_{ijklm} = \mu + \alpha_i + \beta_j + \gamma_k + \delta_l + (\alpha\beta)_{ij} + (\alpha\gamma)_{ik} + (\alpha\delta)_{il} + (\beta\gamma)_{jk} + (\beta\delta)_{jl} + (\gamma\delta)_{kl} \\ + (\alpha\beta\gamma)_{ijk} + (\alpha\beta\delta)_{ijl} + (\alpha\gamma\delta)_{ikl} + (\beta\gamma\delta)_{jkl} + (\alpha\beta\gamma\delta)_{ijkl} + \varepsilon_{ijklm},$$

where $i, j, k, l \in \{0, 1\}$, $m \in \{1, 2\}$, and $\varepsilon_{ijklm} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$. The total number of parameters (excluding σ^2) is:

$$1 + \binom{4}{1} + \binom{4}{2} + \binom{4}{3} + \binom{4}{4} = 1 + 4 + 6 + 4 + 1 = 16.$$

In practice, the three-factor and four-factor interaction terms are assumed negligible under the **Effect Hierarchy Principle**, which states that lower-order effects are more likely to be active and larger in magnitude than higher-order ones.

7 Full Fourth-Order Model

Treatment Combinations and Standard Notation

In the 2^4 factorial design, the 16 treatment combinations are listed in **standard order** (A (Method), B (Time), C (Difficulty), D (Mental State)) is:

$$(1), a, b, ab, c, ac, bc, abc, d, ad, bd, abd, cd, acd, bcd, abcd.$$

Effect Estimates

Let $\mathbf{y}_0 = (y_{(1)}, y_a, y_b, y_{ab}, y_c, y_{ac}, y_{bc}, y_{abc}, y_d, y_{ad}, y_{bd}, y_{abd}, y_{cd}, y_{acd}, y_{bcd}, y_{abcd})^T$ denote the 16×1 vector of **total response values** at each treatment combination (i.e., the sum of the 2 replicate observations at each cell). Let $r = 2$ be the number of replications, $n = 4$ be the number of factors, and $G = \sum_{s=1}^{32} y_s$ be the grand total.

For any effect X , let ℓ_X denote the 16×1 vector of $+1$ and -1 signs for that effect from the Yates table. Then the **treatment contrast** and **effect estimate** are:

$$[X] = \ell_X^T \mathbf{y}_0, \quad \hat{X} = \frac{\ell_X^T \mathbf{y}_0}{r \cdot 2^{n-1}} = \frac{\ell_X^T \mathbf{y}_0}{2 \cdot 8} = \frac{[X]}{16}.$$

General Mean:

$$\widehat{M} = \frac{G}{r \cdot 2^n} = \frac{G}{32}.$$

Main Effects:

$$\begin{aligned} \hat{A} &= \frac{1}{16} [-(1) + a - b + ab - c + ac - bc + abc - d + ad - bd + abd - cd + acd - bcd + abcd], \\ \hat{B} &= \frac{1}{16} [-(1) - a + b + ab - c - ac + bc + abc - d - ad + bd + abd - cd - acd + bcd + abcd], \\ \hat{C} &= \frac{1}{16} [-(1) - a - b - ab + c + ac + bc + abc - d - ad - bd - abd + cd + acd + bcd + abcd], \\ \hat{D} &= \frac{1}{16} [-(1) - a - b - ab - c - ac - bc - abc + d + ad + bd + abd + cd + acd + bcd + abcd]. \end{aligned}$$

Similarly we can have the 2-factor, 3-factor, 4-factor interaction terms.

Sums of Squares

For a linear parametric function $\ell_X^T \mathbf{y}_0$, the general formula for the sum of squares with r replications is:

$$SS_X = \frac{(\ell_X^T \mathbf{y}_0)^2}{r \ell_X^T \ell_X} = \frac{[X]^2}{r \cdot 2^n} = \frac{[X]^2}{32},$$

since $\ell_X^T \ell_X = 2^n = 16$ for all effects in a 2^n design, and $r = 2$. The corrected total and all individual sums of squares are:

$$SS_T = \sum_{i=0}^1 \sum_{j=0}^1 \sum_{k=0}^1 \sum_{l=0}^1 \sum_{m=1}^2 y_{ijklm}^2 - \frac{G^2}{32}, \quad \text{df}_T = 31.$$

Main effect SS (one for each factor, $\text{df} = 1$ each):

$$SS_A = \frac{[A]^2}{32}, \quad SS_B = \frac{[B]^2}{32}, \quad SS_C = \frac{[C]^2}{32}, \quad SS_D = \frac{[D]^2}{32}.$$

Two-factor interaction SS (df = 1 each):

$$SS_{AB} = \frac{[AB]^2}{32}, \quad SS_{AC} = \frac{[AC]^2}{32}, \quad SS_{AD} = \frac{[AD]^2}{32}, \quad SS_{BC} = \frac{[BC]^2}{32}, \quad SS_{BD} = \frac{[BD]^2}{32}, \quad SS_{CD} = \frac{[CD]^2}{32}.$$

Three-factor interaction SS (df = 1 each):

$$SS_{ABC} = \frac{[ABC]^2}{32}, \quad SS_{ABD} = \frac{[ABD]^2}{32}, \quad SS_{ACD} = \frac{[ACD]^2}{32}, \quad SS_{BCD} = \frac{[BCD]^2}{32}.$$

Four-factor interaction SS (df = 1):

$$SS_{ABCD} = \frac{[ABCD]^2}{32}.$$

Pure Error SS (df = $2^n(r-1) = 16 \times 1 = 16$):

$$SS_E = \sum_{i=0}^1 \sum_{j=0}^1 \sum_{k=0}^1 \sum_{l=0}^1 \sum_{m=1}^2 (y_{ijklm} - \bar{y}_{ijkl\bullet})^2 = SS_T - \sum_{\text{all 15 effects}} SS_X.$$

The decomposition identity is:

$$SS_T = SS_A + SS_B + SS_C + SS_D + SS_{AB} + SS_{AC} + SS_{AD} + SS_{BC} + SS_{BD} + SS_{CD} \\ + SS_{ABC} + SS_{ABD} + SS_{ACD} + SS_{BCD} + SS_{ABCD} + SS_E.$$

8 Experiment Results

8.1 Experiment Design

The units are numbered as 1, 2,..., 32. Then executing the code we got the following design.

```
> print(randomized_design)
  Method Time Difficulty MentalState Replication
1 Writing Slow      Easy      Fresh           2
2 Writing Slow      Hard      Fresh           1
3 Writing Slow      Easy      Fresh           1
4 Writing Slow      Easy      Tired          2
5 Typing Slow       Easy      Tired          1
6 Typing Fast       Easy      Fresh           1
7 Typing Slow       Easy      Fresh           2
8 Typing Fast       Hard      Tired          2
9 Writing Fast       Hard      Tired          1
10 Writing Fast      Hard      Fresh           1
11 Writing Fast      Easy      Tired          2
12 Writing Fast      Hard      Tired          2
13 Typing Slow       Hard      Tired          1
14 Typing Slow       Easy      Tired          2
15 Typing Fast       Hard      Fresh           2
16 Writing Fast      Easy      Fresh           1
17 Writing Slow      Easy      Tired          1
18 Typing Fast       Hard      Tired          1
19 Writing Fast      Easy      Tired          1
20 Typing Slow       Hard      Fresh           2
21 Writing Slow      Hard      Fresh           2
22 Typing Fast       Easy      Tired          1
23 Writing Slow      Hard      Tired          1
24 Typing Slow       Easy      Fresh           1
25 Typing Fast       Easy      Tired          2
26 Writing Slow      Hard      Tired          2
27 Writing Fast      Easy      Fresh           2
28 Typing Slow       Hard      Tired          2
29 Typing Slow       Hard      Fresh           1
30 Typing Fast       Hard      Fresh           1
31 Typing Fast       Easy      Fresh           2
32 Writing Fast      Hard      Fresh           2
```

The lists of words of both easy and hard category are numbered as 1, 2,..., 16. We have done the permutation of this set to get the sequence of the lists, which will be used in the experiment.

```

> # Easy words list assignment
> # Lists are numbered from 1,2,...,16
> perm <- sample(1:16)
> print(perm)
[1] 6 10 8 4 13 16 2 5 12 15 7 3 14 9 1 11
> # Difficult words list assignment
> # Lists are numbered from 1,2,...,16
> perm <- sample(1:16)
> print(perm)
[1] 10 2 6 14 15 4 3 13 5 12 9 7 8 1 16 11

```

So, 1st experimental unit with category - easy words will get 6th easy word list, 2nd will get 10th easy word list and so on. 1st experimental unit with category - difficult words will get 10th difficult word list, 2nd will get 2nd difficult word list and so on.

8.2 Collection of data

Word List	Treatment	Count	Word List	Treatment	Count
H6	W/F/H/T	15/20	E1	W/F/E/F	9/20
H4	T/S/H/T	19/20	H7	W/S/H/T	15/20
E15	W/S/E/T	16/20	E8	W/S/E/T	15/20
H2	T/F/H/T	9/20	E7	W/F/E/T	9/20
H12	W/S/H/F	16/20	E3	T/F/E/T	11/20
H3	T/F/H/F	11/20	H13	T/F/H/T	12/20
E6	W/S/E/F	16/20	E11	T/F/E/F	19/20
H1	T/S/H/F	9/20	E12	W/F/E/F	11/20
H11	W/F/H/F	14/20	E14	T/S/E/F	18/20
H13	T/F/E/F	10/20	E10	W/S/E/F	17/20
H10	W/S/H/F	17/20	E16	T/S/E/F	16/20
E1	W/F/E/F	9/20	H5	T/S/H/F	13/20
H7	W/S/H/T	15/20	H16	T/F/H/F	10/20
E8	W/S/E/T	15/20	H14	W/F/H/F	13/20
E7	W/F/E/T	9/20	E9	T/F/E/T	12/20
E3	T/F/E/T	11/20	E5	T/S/E/T	18/20

Treatment code: W = Writing, T = Typing; S = Slow, F = Fast; E = Easy, H = Hard; F = Fresh, T = Tired.

9 Experiment Results and analysis

The experiment was conducted with $N = 32$ participants assigned to the 2^4 full factorial design with 2 replications. Two nested ANOVA models were fitted to the response variable (number of words recalled out of 20): a **second-order model** retaining only main effects and two-factor interactions, and the **full saturated fourth-order model** retaining all effects up to the four-factor interaction. The R output for both models is presented and interpreted below.

9.1 Fitting the Second-Order Model

The 3- and 4-factor interaction terms are assumed negligible and pooled into the residual, giving an effective error with $df_{\text{Res}} = 21$ and $MS_{\text{Res}} = 5.14$.

Source	df	SS	MS	F	p-value	
Method (A)	1	0.03	0.03	0.006	0.93858	
Time (B)	1	140.28	140.28	27.301	3.53×10^{-5}	***
Difficulty (C)	1	0.28	0.28	0.055	0.81729	
Mental State (D)	1	1.53	1.53	0.298	0.59089	
Method $\times$ Time (AB)	1	0.03	0.03	0.006	0.93858	
Method $\times$ Difficulty (AC)	1	42.78	42.78	8.326	0.00885	**
Method $\times$ Mental State (AD)	1	5.28	5.28	1.028	0.32221	
Time $\times$ Difficulty (BC)	1	9.03	9.03	1.758	0.19917	
Time $\times$ Mental State (BD)	1	9.03	9.03	1.758	0.19917	
Difficulty $\times$ Mental State (CD)	1	16.53	16.53	3.217	0.08728	.
Residuals	21	107.91	5.14			
Total	31					

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Key Interpretations of the Second-Order Model

- **Time (B) — Highly Significant** ($p = 3.53 \times 10^{-5}$): The dominant effect in the experiment ($F_{1,21} = 27.301$). A slower transcription pace leads to significantly higher word recall. This aligns with the *cognitive load hypothesis*, as taking more time allows for better elaborative encoding and rehearsal.
- **Method $\times$ Difficulty Interaction (AC) — Significant** ($p = 0.00885$): The effect of the transcription method (writing vs. typing) depends heavily on the difficulty of the words. Handwriting likely provides a recall advantage specifically for hard words because the slower process forces deeper cognitive processing.
- **Difficulty $\times$ Mental State (CD) — Marginal** ($p = 0.08728$): A borderline interaction suggesting that being mentally tired while tackling hard words compounds to impair recall less than either factor alone. This makes biological sense (due to depleted working memory) and might cross the significance threshold in a higher-powered study.
- **Remaining Effects — Not Significant:** Main effects for Method (A), Difficulty (C), Mental State (D), and all remaining interactions were not significant ($p > 0.19$). Notably, while the transcription Method isn't significant on its own, it remains crucial when paired with word Difficulty (the AC interaction).

9.2 Fitting the Full Fourth-Order Model

The full saturated model includes all $2^4 - 1 = 15$ factorial effects. The 3- and 4-factor interaction terms are *not* pooled into the error here; instead the pure replication error with $df_E = 16$ and $MS_E = SS_E/16 = 62.50/16 = 3.91$ serves as the denominator for all F -tests.

Source	df	SS	MS	F	p-value	
Method (A)	1	0.03	0.03	0.008	0.92984	
Time (B)	1	140.28	140.28	35.912	1.88×10^{-5}	***
Difficulty (C)	1	0.28	0.28	0.072	0.79188	
Mental State (D)	1	1.53	1.53	0.392	0.54008	
Method $\times$ Time (AB)	1	0.03	0.03	0.008	0.92984	
Method $\times$ Difficulty (AC)	1	42.78	42.78	10.952	0.00443	**
Method $\times$ Mental State (AD)	1	5.28	5.28	1.352	0.26198	
Time $\times$ Difficulty (BC)	1	9.03	9.03	2.312	0.14789	
Time $\times$ Mental State (BD)	1	9.03	9.03	2.312	0.14789	
Difficulty $\times$ Mental State (CD)	1	16.53	16.53	4.232	0.05635	.
Method $\times$ Time $\times$ Difficulty (ABC)	1	22.78	22.78	5.832	0.02807	*
Method $\times$ Time $\times$ Mental State (ABD)	1	9.03	9.03	2.312	0.14789	
Method $\times$ Difficulty $\times$ Mental State (ACD)	1	0.78	0.78	0.200	0.66072	
Time $\times$ Difficulty $\times$ Mental State (BCD)	1	3.78	3.78	0.968	0.33982	
Method $\times$ Time $\times$ Difficulty $\times$ Mental State (ABCD)	1	9.03	9.03	2.312	0.14789	
Pure Error	16	62.50	3.91			
Total	31					

Key Interpretations of the Full Fourth-Order Model

- **Model Comparison:** By using pure replication error ($MS_E = 3.91$, $df = 16$), the full model provides a more conservative and precise test, resulting in uniformly larger F -values than the second-order model.
- **Time (B) — Highly Significant** ($p = 1.88 \times 10^{-5}$): Remains the overwhelmingly strongest effect. Slow transcription consistently and substantially yields better word recall than fast transcription.
- **Method $\times$ Difficulty (AC) — Significant** ($p = 0.00443$): Strengthens the earlier finding with greater confidence: the memory retention benefit of writing versus typing depends critically on whether the material is easy or hard.
- **Method $\times$ Time $\times$ Difficulty (ABC) — Significant** ($p = 0.02807$): A **new finding** indicating the AC interaction varies by pace. For example, the recall benefit of handwriting for hard words is more during slow transcription; at a fast pace, the benefit of handwriting may not be much significant.
- **Difficulty $\times$ Mental State (CD) — Marginal** ($p = 0.05635$): Remains borderline. The interaction effect of word difficulty and mental fatigue is still statistically inconclusive at $\alpha = 0.05$, but this time has lesser p -value.
- **Remaining Higher-Order Interactions — Not Significant** ($p > 0.14$): All other 3-way and 4-way interactions are comparable to pure error. This validates the *Effect Hierarchy Principle*, confirming that these higher-order effects are mostly noise.

9.3 Summary and Comparison of Both Models

Effect	SS	2nd-Order p	Full Model p	Verdict
B (Time)	140.28	3.53×10^{-5} ***	1.88×10^{-5} ***	Significant
AC (Method $\times$ Difficulty)	42.78	0.009 **	0.004 **	Significant
ABC (Method $\times$ Time $\times$ Difficulty)	22.78	-	0.028 *	Significant (full only)
CD (Difficulty $\times$ Mental State)	16.53	0.087	0.056	Marginal
All others	≤ 9.03	> 0.19	> 0.14	Non-significant

The two models are broadly consistent: both identify B and AC as the primary drivers of recall variability. The full model additionally reveals the significant ABC three-factor interaction, which the second-order model would have missed by design. The two models together paint a coherent picture: **the pace of transcription is the single most powerful determinant of word recall**, and **the advantage of handwriting over typing is real but contingent on both word difficulty and transcription speed**. The main effect of method alone, taken in isolation across all conditions, is negligibly small.

10 Conclusion: Writing vs. Typing for Memory Retention

The experiment reveals that there is no single best method; instead, memory retention is governed by a hierarchy of factors and interactions.

- **Dominance of Pace (Time):** The single most powerful determinant of recall is the pace of transcription. Slower transcription consistently leads to better recall by allowing more time for semantic processing and rehearsal.
- **Conditional Advantage of Handwriting:** The method of transcription (writing vs. typing) has no significant main effect ($p > 0.93$). However, a significant $Method \times Difficulty$ interaction ($p = 0.004$) shows that handwriting provides a recall benefit **only** when word lists are difficult.
- **The Three-Factor Constraint (ABC):** The writing advantage for difficult words is further moderated by time ($p = 0.028$). The benefit of handwriting for hard words disappears if the pace is too fast, as while writing under time pressure degrades the cognitive encoding benefit.
- **Negligible Factors:** Neither Mental State nor word Difficulty alone were significant main effects ($p > 0.54$). There is only a borderline interaction ($p \approx 0.056$) suggesting fatigue may compound the difficulty of unfamiliar words.

Final Suggestion

To maximize memory retention: **work slowly**. Handwriting is most effective for difficult material when performed at a deliberate, unhurried pace. Under easy conditions or fast-paced requirements, typing and writing yield comparable results.

11 R Codes

```
1 # 1. Define the factors and their levels
2 factors <- list(
3   Method = c("Writing", "Typing"),
4   Time = c("Slow", "Fast"),
5   Difficulty = c("Easy", "Hard"),
6   MentalState = c("Fresh", "Tired")
7 )
8
9 # 2. Generate the 16-run full factorial base
10 basic_design <- expand.grid(factors)
11
12 # 3. Create 2 replications
13 rep1 <- basic_design
14 rep1$Replication <- 1
15 rep2 <- basic_design
16 rep2$Replication <- 2
17
18 # Combine into a single dataframe of 32 units
19 full_design <- rbind(rep1, rep2)
20
21 # 4. Randomize the treatment assignments
22 randomized_design <- full_design[sample(nrow(full_design)), ]
23 rownames(randomized_design) <- NULL # Reset row numbering
24
25 # Easy words list assignment (Randomized)
26 perm_easy <- sample(1:16)
27
28 # Difficult words list assignment (Randomized)
29 perm_hard <- sample(1:16)
30
31 # Response variable vector (Word Count)
32 # Ordered according to the 32-run randomized design
33 y <- c(17, 16, 16, 15, 18, 10, 16, 12, 15, 14,
34       10, 14, 19, 17, 10, 9, 16, 9, 9, 13,
35       17, 11, 15, 18, 12, 16, 11, 18, 9, 11,
36       19, 13)
37
38 # Add the vector in the dataframe
39 randomized_design$Words_Remembered <- y
40
41 # Fit the Second-Order Model (Main Effects + 2-Way Interactions)
42 model_second_order <- aov(
43   Words_Remembered ~ (Method + Time + Difficulty + MentalState)^2,
44   data = randomized_design
45 )
46 summary(model_second_order)
47
48 # Fit the Full Fourth-Order Model
49 model_full_saturated <- aov(
50   Words_Remembered ~ Method * Time * Difficulty * MentalState,
51   data = randomized_design
52 )
53 summary(model_full_saturated)
```

Listing 1: R Script for Experimental Design and ANOVA Analysis

12 Appendix: Experimental Word Lists

This appendix contains the 32 randomized word lists used in the memory retention experiment. Each list consists of 20 words categorized by difficulty.

12.1 Factor Level: Easy Words

List E1: bell, bridge, coin, desk, farm, gold, island, king, market, ocean, plate, rock, shirt, tail, uncle, wall, yard, zoo, apple, boat

List E2: bird, cake, door, fish, gate, home, juice, lamp, milk, nose, pear, road, star, tree, vase, window, bear, cup, egg, frog

List E3: book, chair, duck, fire, girl, hand, ice, jump, leaf, moon, nest, park, rain, shoe, tent, wing, ant, car, dog, gift

List E4: ball, corn, ear, flag, goat, hill, jam, knee, lion, mountain, night, pen, ring, sun, town, wolf, bed, cat, eye, hat

List E5: box, cloud, doll, food, glass, horse, jar, kite, map, net, owl, pig, rose, salt, ship, toy, wood, bag, cow, fan

List E6: boy, coat, deer, foot, game, house, jaw, land, mouth, nurse, orange, pot, seed, soap, train, watch, zoo, arm, desk, girl

List E7: bus, desk, face, garden, hat, ink, key, leg, money, nut, paint, queen, sand, snow, truck, wheel, ball, duck, egg, fork

List E8: cap, day, feel, grass, hair, iron, kid, lock, moon, oil, pipe, roof, shell, spoon, unit, wave, ant, bird, cake, door

List E9: city, dog, fruit, heart, jump, lake, meat, page, rain, sky, tool, vest, wind, apple, box, car, ear, fish, goat

List E10: cup, eye, fire, gold, hill, ice, king, leaf, milk, nose, pear, rock, star, tree, wall, yard, bear, bird, chair, door

List E11: fan, frog, gift, hand, home, jam, knee, lamp, map, nest, park, ring, shoe, town, vase, wing, book, cat, desk, egg

List E12: fork, gate, hat, horse, jar, kite, lion, moon, night, owl, pen, rose, ship, toy, wheel, wolf, ball, boy, cloud, dog

List E13: game, girl, house, ink, jaw, lake, mouth, nut, paint, pig, salt, soap, tent, unit, watch, wood, arm, box, cake, duck

List E14: garden, glass, head, iron, key, leg, money, net, orange, pot, sand, spoon, train, vest, wind, zoo, bag, car, deer, eye

List E15: grass, hair, ice, jump, king, lock, meat, oil, page, queen, roof, shell, sun, truck, wave, yard, bell, bus, cup, fish

List E16: heart, hill, ink, jaw, kite, lamp, moon, nose, park, pipe, ring, seed, sky, tool, vase, wing, apple, bed, coin, dog

12.2 Factor Level: Hard Words

List H1: analyze, benefit, complex, derive, evaluate, factor, global, hypothesis, identify, journal, knowledge, logic, method, notion, objective, perspective, quality, research, status, theory

List H2: abstract, bias, capacity, data, energy, finance, generate, hierarchy, impact, justify, labor, mental, network, option, phase, radical, source, target, ultimate, valid

List H3: acquire, brief, category, design, element, format, grade, instance, link, major, normal, output, period, range, sector, task, unique, vary, welfare, zone

List H4: adapt, bond, clause, device, ethnic, framework, goal, image, item, layer, margin, occupy, policy, register, shift, trend, utility, volume, whereas, academy

List H5: adult, bulk, code, discrete, estate, function, grant, index, job, levy, media, occur, prime, rely, site, transfer, vehicle, vision, waive, adjacent

List H6: advocate, channel, core, edit, expand, gender, hence, inhibit, label, mode, native, orient, prior, remove, solo, thesis, underlying, visible, warrant, aggregate

List H7: alter, civil, credit, emerge, expert, graph, ideal, input, locate, motive, negate, parade, quote, robust, sphere, trace, unique, verify, wisdom, allocate

List H8: amend, classic, cycle, enable, export, guide, ignore, internal, logic, mature, neutral, passive, ratio, scheme, stable, transit, unit, virtual, yield, ambiguous

List H9: aspect, commit, deduce, enforce, expose, huge, imply, invest, lucid, medical, notable, precise, regime, scope, survey, transport, utilize, visual, x-ray, analogy

List H10: assign, compute, deny, entity, external, idler, intense, joint, luxury, modify, novel, predict, relief, sequence, sustain, trigger, vacuum, volume, yield, analyze

List H11: assist, conduct, detect, error, facility, ignore, intent, junior, maintain, monitor, nuclear, promote, retain, series, symbol, ultimate, vary, wealth, youth, benefit

List H12: assume, consult, deviate, evolve, federal, impart, interact, kernel, maximize, neglect, obtain, pursue, reveal, severe, target, undergo, vast, weight, zenith, complex

List H13: attach, consume, display, exceed, finite, impose, invade, latent, migrate, notify, output, quality, reward, signal, tenure, unique, vital, welfare, zone, derive

List H14: attain, contact, diverse, exclude, flexible, indeed, invite, legacy, minimal, obey, parallel, radical, rhythm, stress, theme, utility, vivid, width, zero, evaluate

List H15: author, convert, draft, exhibit, focus, indoor, island, legend, minor, obtain, partner, reform, ritual, submit, topic, valid, volume, worth, yield, factor

List H16: aware, create, dynamic, expand, formal, induce, item, local, modify, occur, permit, reject, sample, survey, theory, value, vision, weight, youth, global